

Platinum

△ Piece of platinum in its natural state

When Spanish conquistadors in Columbia first found platinum in the 16th century, they called it platina, meaning “little silver”. To them it was worthless, a distraction in their search for gold. Today, however, it is one of the most precious metals on Earth, both due to its scarcity and its properties as a catalyst, speeding up chemical reactions while remaining inactive itself. As well as its use in fine jewellery, it is a vital component in converting crude oil into petroleum, and also plays a role in reducing pollution from cars as a form of filter (see box below).

Specification

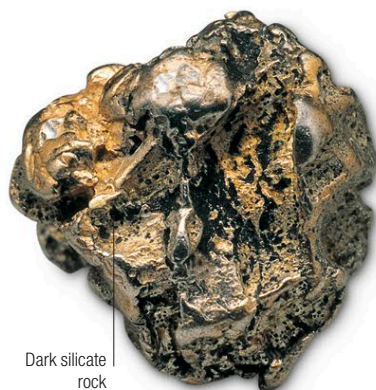
Chemical name Platinum | **Formula** Pt | **Colours** White, silver grey, steel-grey | **Structure** Isometric (cubic) | **Hardness** 3.5 | **SG** 21.45 | **RI** 2.19 | **Lustre** Metallic | **Streak** n/a
Locations South Africa, Russia, Canada

Rough



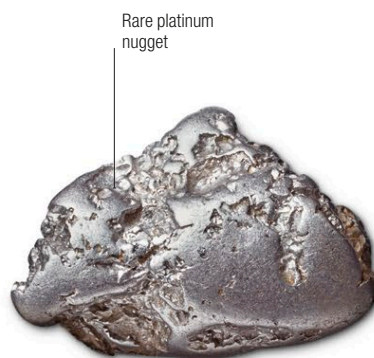
Grains vary in size

Grains of platinum | Platinum grains occur naturally but usually include traces of other metals including iron, palladium, rhodium, and iridium.



Dark silicate rock

Embedded in rock | When platinum is found as grains, flakes or thin layers in silicate rock, as in this example, it is typically mixed with other minerals and has to be separated out.



Rare platinum nugget

Platinum nugget | Although most platinum occurs naturally as grains, it is only rarely found in nugget form. Platinum nuggets such as this do not tarnish.

Settings



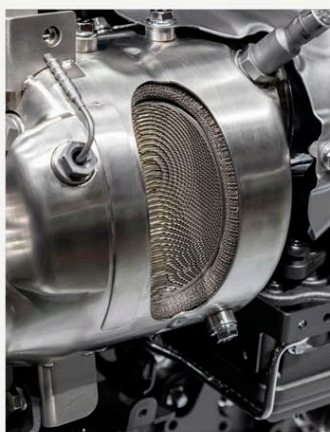
Channel-set diamonds

Eternity ring | Jewellery firm De Beers invented the idea of the diamond eternity ring in the 1960s – the platinum version is the most valuable of their range.

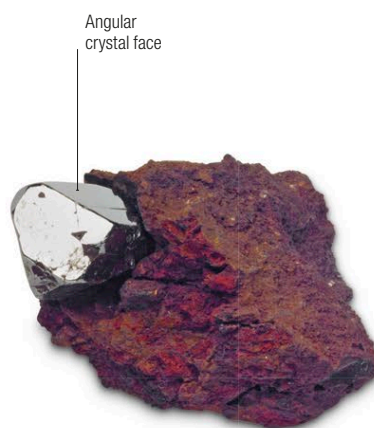
Reducing gas

Catalytic converters

Platinum reduces pollutants from car engines by converting poisonous gases into less harmful substances. Since 1974, when the USA introduced new laws on air quality, catalytic converters in vehicles have become a worldwide phenomenon. Catalytic converters use platinum to minimize the emission of noxious gas from engines – the platinum catalyst rips apart the toxic nitrogen dioxide and allows the molecules to re-form in less toxic combinations.



Platinum at work This cross-section of a catalytic converter shows the platinum grains it contains.



Angular crystal face

Sperrylite crystal | Sperrylite is a compound of platinum and arsenic, and is valued by collectors as a specimen rather than an ore of platinum.

Traces of platinum from around 1200 BCE have been found in ancient Egyptian tombs